

Sunil Choudhary

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Profile

AI and computer vision researcher working on digital twins for industrial environments. PhD defence planned for February 2027, available from March 2027.

Research Interests

Digital twins ▪ Synthetic data ▪ Sim-to-real transfer ▪ 3D reconstruction ▪ 6D pose estimation ▪ Radiance fields ▪ Structure-from-Motion ▪ Industrial robotics and datasets ▪ Domain adaptation

Education

École Nationale Supérieure d'Arts et Métiers (ENSAM ParisTech) — **Paris, France**
PhD candidate in Computer Science — *March 2024 – March 2027*
- Thesis title: "6D Camera Pose Refinement and 3D Reconstruction via a Radiance Field-Based Approach: an industrial perspective"
- Thesis director: Yohan Dupuis; Co-director: Vincent Havard; Co-supervisor: Nicolas Ragot
- Doctoral school ED SMI 432; defence planned for February 2027

Université Bourgogne Europe — **Dijon, France**
Master in Artificial Intelligence (M2 Santé-AI) — *September 2023*
- Thesis title: "3D Orientation Estimation of a 6 DoF Optical Microrobot using Deep Learning"
- Thesis internship done at ISIR, Sorbonne University, Paris, France
- Programme covering AI, cloud computing, image and signal processing, and cybersecurity topics

Université Marie & Louis Pasteur — **Dijon, France**
Master in Mathematical Physics (M2 Math4Phys) — *July 2020*
- Programme focused on advanced mathematics courses and their applications in physics

Indian Institute of Science Education and Research (IISER) — **Pune, India**
Four-year Bachelor of Science (BS) — *July 2019*
- Programme focused on the fundamental sciences, with a specialisation in Mathematics

Work Experience

CESI LINEACT Laboratory (UR 7527) — **Rouen, France**
Doctoral Researcher, Computer Vision, AI and Robotics — *March 2024 – March 2027*
- Funded by **FUSION**, *Framework for Universal Software Integration in OpeN robotics*, awarded through the "i-Démo régionalisé" call of the French government and the Normandy region under France 2030, with Assystem, Orano and Conscience Robotics as industrial partners. FUSION aims to democratise robotics by integrating AI and eXtended Reality; within it I tackle 3D reconstruction and 6D camera pose estimation in industrial environments.
- Collaborator for CESI on **REMOTED**, *Robotic EnvironMent hybridizatiOn rendering based-on data-efficient radiance field TEchnologies and Digital twin* (PHC Balaton 52934RC, Campus France), a bilateral French–Hungarian collaboration with ELTE University, Budapest, on a hybrid, data-efficient 3D reconstruction pipeline for industrial environments where real data is scarce. Two research visits to Budapest, which converted into collaborative projects and exchanges.
- Built and publicly released the first paired physical/digital-twin industrial RGB-D dataset, captured with a stereo camera and a 16-camera OptiTrack MoCap rig for millimetre-accurate ground truth, then replayed along the identical trajectory in a digital twin of the environment in Unity [5].
- Benchmarked Structure-from-Motion backends against Gaussian Splatting variants across the real and synthetic domains on pose accuracy, rendering fidelity and runtime; quantified the sim-to-real gap, showed that method rankings transfer between domains, measured the robustness of Gaussian Splatting on an industrial dataset, and developed a hybridization technique that narrows the gap by increasing synthetic data and reducing the real data required [3, 4].
- Met regularly with project partners, presented demonstrations to funders, planned experiments, and presented this work at seminars and international conferences.

CESI School of Engineering — **Rouen, France**
Course Instructor and Research Internship Supervisor — *Jan 2026 – Aug 2026*

- **Deep Learning and AI Ethics** (45 hours, second-year master's students) and **Machine Learning and AI** (15 hours, fourth-year engineering students): workshops on neural network architectures, regression and classification models and hyperparameter tuning, with applications to medical data and a discussion of AI ethics and research integrity; project sessions, follow-up and defence evaluation including the students' ethical analysis.
- **Master's thesis internship** (1 student, April – August 2026), "Hybrid Reconstruction using Real and Synthetic Images of Industrial Objects via 3D Gaussian Splatting"; showed how far CAD-derived synthetic views can replace real acquisitions.
- **Master's group project** (3 students, Jan – April 2026), "Benchmarking 3D Gaussian Splatting Techniques in Industry and Its Real Use Cases".

ISIR, Sorbonne University **Paris, France**

Robotics Engineer Intern (M2 thesis) *March 2023 – Nov 2023*

- Supervisors: Sinan Haliyo, Ferhat Sadak and Edison Gerena. Worked on microrobot datasets and 3D orientation estimation, with applications to in vitro fertilisation (IVF).
- Fine-tuned YOLOv7 and DeepSORT for microrobot and trapping-point tracking and built an EfficientNet feature extractor with custom CNN and feature-fusion layers, showing that the full 3D orientation of a microrobot and its trapping points can be recovered from the microscope image stream alone, which makes closed-loop tracking during manipulation feasible. Published in *Robotica* [1]; this work fed into an early stage of **MovaLife**, a deep-tech microrobotics company founded by Edison Gerena, its CTO.

Research Contributions

[1] Journal: **Sunil Choudhary**, Ferhat Sadak, Edison Gerena and Sinan Haliyo. "Three-dimensional optical microrobot orientation estimation and tracking using deep learning." **Robotica**, 43(2), 616–637, Cambridge University Press, 2025. **Published**. doi:10.1017/S0263574724002091

[2] Conference: **Sunil Choudhary**, Nicolas Ragot, Vincent Havard and Yohan Dupuis. "Evaluating Robustness of 3D Gaussian Splatting-Based 6D Camera Pose Refinement Under Degraded Conditions for Lightly Textured Industrial Synthetic Objects." **IECON 2025**, 51st Annual Conference of the IEEE Industrial Electronics Society, Madrid, Spain, 2025. **Published**. doi:10.1109/IECON58223.2025.11221686

[3] Conference: **Sunil Choudhary**, Mickyas Tamiru Asfaw, Aristide Laignel, Nicolas Ragot, Vincent Havard, Kristian Fenech and Yohan Dupuis. "Mind the Domain Gap: Benchmarking Structure-from-Motion Methods on Paired Industrial Physical-Digital Twin Datasets to Enable Radiance Field Rendering." **ICPR 2026**, 28th International Conference on Pattern Recognition, MVI2 Workshop, Lyon, France. **Published**. hal-05677254

[4] Journal: **Sunil Choudhary**, Mickyas Tamiru Asfaw, Nicolas Ragot, Kristian Fenech, Aristide Laignel, Aron Fothi, Gabor Baranyi, Vincent Havard and Yohan Dupuis. "Exploring the Physical and Digital Domain Gap in 3D Reconstruction: A Benchmark of Radiance Field Rendering Techniques on Paired Industrial Twins." **Array** (Elsevier), 2026. **Under review**.

[5] Dataset: **Sunil Choudhary**, Mickyas Tamiru Asfaw, Aristide Laignel, Nicolas Ragot, Vincent Havard, Kristian Fenech and Yohan Dupuis. "Physical and Digital Twin Industrial RGBD Dataset with same camera trajectory for 3D vision applications." **Zenodo**, v1, June 2026. **Published**. doi:10.5281/zenodo.20830059

Links: Project page ▪ Dataset ▪ Code

Reviewer: **Automation in Construction** (Elsevier) ▪ **Intelligent Robotics Fair 2026**, Budapest, Hungary ▪ **ICPR 2026**, 28th International Conference on Pattern Recognition, Lyon, France

Technical Skills

Programming: Python, MATLAB, Bash, LaTeX, C#, C, C++

Libraries: COLMAP, Open3D, OpenCV, PyTorch, Scikit-Learn, Optuna, NumPy, SciPy, Pandas, Matplotlib

Robotics & simulation: NVIDIA Isaac Sim, Unity, Blender, OptiTrack MoCap, ROS 2, stereo cameras (active and passive)

Infrastructure: Linux, Docker, Apptainer, Slurm (GPU/HPC), Git, GitHub, GitLab CI

Languages: English (fluent), Hindi (native), Rajasthani (native), French (A1)

Talks, Seminars and Research Visits

July 2026: "My thesis in 3 minutes" challenge, CESI LINEACT seminar, Toulouse, France.

June 2026: REMOTED project, research visit to ELTE University (one week), Budapest, Hungary.

June 2026: Intelligent Robotics Fair 2026, Bosch Innovation Campus, Budapest, Hungary.

January 2026: Poster and oral presentation, CESI LINEACT seminar, Dijon, France.

October 2025: Paper and poster presentation, IEEE IECON 2025, Madrid, Spain.

October 2025: Research talk on behalf of CESI, NIPG Seminar, ELTE University, Budapest, Hungary.

October 2025: REMOTED project, research visit to ELTE University (two weeks), Budapest, Hungary.

June 2025: CVPR@Paris, a one-day event with poster presentations of CVPR 2025 papers, Paris, France.

May 2025: Days of the Doctoral School, second-year session, ENSAM and CNAM, Paris, France.

May 2025: Poster presentation, CESI LINEACT Unity Seminar, Saint-Nazaire, France.

June 2024: "My thesis in 3 minutes" presentation, CESI LINEACT seminar, Rouen, France.

Fellowships and Awards

2020: **ISITE-BFC Fellowship** for master's study in France (EUR 5,555).

2019: **Junior Research Fellowship (JRF)**, CSIR-NET, All India Rank 149; five-year fellowship to pursue a PhD in Mathematics in India.

2018: **IIT Bombay Research Internship Award**, six-month funded research internship.

2015: **INSPIRE Fellowship**, Government of India, awarded to the top 1% of candidates nationally for undergraduate study in the natural and basic sciences.

2015: **JEE-Advanced**, All India Rank 3181 among 1.35 million candidates.

References

Yohan Dupuis: Director of Research and Development, CESI, Paris, France (PhD thesis director).
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Vincent Havard: Head of the Department of AI and Trustworthy Digital Systems, CESI, Rouen, France (PhD co-director).
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Nicolas Ragot: Associate Professor, CESI, Rouen, France (PhD co-supervisor).
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Kristian Fenech: Assistant Professor, Eötvös Loránd University (ELTE), Budapest, Hungary.
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